A sovereign bond based index for financial instability in the euro zone and the role of carry trade*

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Abstract

The paper suggests a simple Financial Stress Index for the euro zone with three possible different outcomes depending on the severity of the crisis. This indicator is based on the specific shape of the credit term structure for the sovereign bond market of the two largest peripheral countries in the euro area. The paper discusses how some key features of the term structure are linked to government debt carry trade. In order to assess the performance of the proposed market-based indicator, the paper shows how the identified episodes of financial turmoil are related to the timing and to the intensity of unconventional measures in the area.

Keywords: Financial stress index, Unconventional Measures, Asset Swap, EONIA, Term structure of credit spread, Peripheral euro government bonds, Government debt carry-trade.

JEL Classification: C51, E58, E65, G12, G18, H63.

1 Introduction

The crisis begun in 2007 has increased the need for a simple indicator of financial instability, that should both i) identify the episodes of systemic stress and ii) display the level of instability. The signal produced by an indicator of financial instability is called Financial Stress Index (FSI); it has aroused a significant interest in recent literature (see e.g. [CEL11; ECB10a; HM05; IL06; NP05]) and even in private sector research (see e.g. BCA Research that reports a monthly FSI for U.S.A.), since a clear identification of systemic risk is the prerequisite for macroprudential supervisory and regulatory measures.
In FSIs the main idea is to build a composite indicator of money, bond (government, corporate and MBS), equity, volatility and foreign exchange markets [CEL11; ECB10a; IL06; NP05], as well as to consider some relevant features of financial intermediaries and market infrastructure; in some cases an FSI can be based on privileged information (e.g. recourse to central bank facilities of systemically relevant intermediaries [ECB10a] or a banking commission's list of institutions that are under special scrutiny [HM05]).

This index can be constructed following two main steps: i) a selection of the components and ii) an aggregation method. As already underlined in the literature, "composite indicators are relatively rough by nature, and thus share specific problems that limit their comparability and interpretability, such as the wide-ranging freedom of choice as to the selection of both the input series and the aggregation method" (see [ECB10a], pg. 141).

Selection should be based on systemic relevance and ability to reflect market participants' behaviour. In [CEL11] selection is based on a uniform set of indicators applied across a panel of 17 economies. More often selection may be performed including in the indicator some key elements relevant for the economy under analysis. For example in [IL06] the commodity market is a relevant component in the FSI for Canada, while in [HM05] the Swiss index is mainly based on the banking system. In this paper we follow this last approach and we therefore include some elements that characterize the euro area economy.

Aggregation is completely arbitrary and there is no *a priori* recipe. Generally it is performed first by constructing sub-indices that aggregate the variables relative to a given market/sector (e.g. the foreign exchange market or the banking sector), and then by averaging out the different sub-indices with different weighting schemes. The choice of the weighting method is crucial, as it drives the impact of each sub-index in the FSI. A quite common weighting method described in the literature is to consider equal-variance-weighting for the different sub-indices (see e.g. [CEL11; HM05; IL06]); however, in this case, variables are assumed to be normally distributed: an unrealistic assumption, especially in crisis periods. Often, the chosen scheme is the one that works better within past dataset (see e.g. [IL06]), however it is a documented fact that FSIs depend strongly on the chosen weighting mix [HM05]. Our approach is rather different because it comes from the following observation: if financial stress is truly systemic, we expect that effects should materialize in several financial sectors at the same time (see e.g. [ECB10a]); and in particular a crisis appears more evident especially in the market characterizing the key financial risks of the economy under analysis. In this paper, we focus on this key market for the specific economy of interest and, within the proposed approach, the aggregation step becomes straightforward.

In this paper we suggest a new FSI for the euro area economy; it is a **simple discrete market-based** indicator where the aggregation step is truly elementary due to some specific characteristics of the euro area economy.

Why a financial instability indicator has to be **simple**? A simple and discrete indicator is particularly suited for a policy maker; as already mentioned in the literature (see e.g. [NP05]), an FSI should not include an as-complete-as-possible financial description because it is not used as a *black box* to determine policy action, but as an indicator of the need of a more detailed quantitative and qualitative analysis.

A similitude could clarify this aspect, which is crucial in our approach. In the medical sector, for instance, a clinical thermometer is surely the simplest indicator of illness. In spite of the complexity of the human body, the temperature measured in some parts of the body by a clinical thermometer provides two simple pieces of information: the presence of some kind of illnesses and their severity, indicated by higher temperatures. The information got by an average user can be coded into three colours: no fever (Green), low temperature – i.e. up to a specified threshold as 39 degrees Celsius – (Yellow), and high temperature – higher than the defined threshold – (Red). The main advantages of a thermometer are evident: it is very simple (everybody is able to use it) and direct (just one physical property is measured). Clearly a thermometer does not say anything about the disease; an assessment of the medical situation can only be completed after both quantitative and qualitative additional analysis conducted by specialists (e.g. doctors, radiologists, etc.).

Similarly, we would like to set up an FSI on a specific economy, that allows to spot the crisis through a single **discrete** signal as simple as

Red:	severe financial market disruption;
Yellow:	dysfunctional financial market;
Green:	no stress situation.

Simplicity is also a consequence, as in the clinical thermometer example, of the measure of just **one single quantity** (a market-based financial quantity in our case); we avoid most of the aggregation arbitrariness deriving from the use of non-uniform quantities.
Last but not least, a fruitful and immediate FSI should be **market-based**. A market-based index could be "calculated daily, if not even more frequently" (see [NP05], pg. 3), as prices are available instantaneously and continuously; prices reflect views of forward looking investors and "react nearly instantaneously to investors' judgements about financial conditions" and to "market participants' attitude toward risk" (see [NP05], pg. 4).

In this study we focus on the euro area economy. The element that characterizes this area is the presence of a monetary but not a fiscal union. The systemic risk that is perceived by the market as the most relevant is a break-up: this key risk has been underlined in the financial press (see e.g. [BD11; Rog13]) and in the literature (see e.g. [BPS14] and references therein). It is definitely related to an event that would not "*preserve the euro*", as ECB President has clearly explained on the 26^{th} of July 2012 [Dra12].
For "large" countries, i.e. the ones that involve a significant fraction of area's GDP (e.g. more than 10%), unsustainability of sovereign debt (i.e. idiosyncratic country credit risk) is strictly related to a break-up (i.e. systemic risk). On one hand, a perceived higher probability of break-up could lead to a sovereign default, since a unique currency allows a quick outflow of capitals from peripheral countries (often called GIPSI, an acronym used to indicate Greece, Ireland, Portugal, Spain and Italy) to core countries, perceived as safer, in order to avoid the devaluation strategy – for the peripheral currency – that would follow a break-up; the well known *flight-to-quality* effect may have as a main consequence the unsustainability of a peripheral sovereign debt leading to a credit event. On the other hand, the reverse is also true: for "large" countries, such as Italy or Spain, a default could lead to the political indefensibility of the euro currency and the need for a break-up.

The indicator we suggest in this paper is simple and directly observable from the government bonds' market. We focus our analysis on the period between the 1^{st} of January 2007 and the 31^{st} of December 2013. We first construct a country-index based on sovereign credit term structure for the two "large" GIPSI countries; the FSI is then defined as the maximum between the values of the two country-indices: it is enough that one of these countries could not refinance its debt in order to induce a currency break up.
Country index is related to the shape of term structure of Asset Swap spread over EONIA for some government bond markets; Asset Swap is a standard derivative contract in credit markets [Sch03] that has been modified in this study in order to refer to what is considered the benchmark risk free curve in euro interbank market, the EONIA curve. The corresponding spread can be observed in each business day and involves all end-of-the-day prices in the two markets (last-trade for government bonds and last-quotes in the interest rate swap market). Asset swap spread resembles the difference between sovereign yield and swap rate with the same maturity, as considered in [BPS14]: on one side, the proposed approach is more precise when constructing an FSI, because a swap rate has a systematic

risk component - due to the well known Euribor *vs* OIS spread - that could not be neglected; on the other side, it involves a derivative - the asset swap - strictly related to the carry trade strategy. Furthermore, in the literature, the spread between sovereign yield and swap rate at a given maturity is generally considered as indicator of systemic risk (e.g. 5 years in [BPS14]): as we show in this study, the most relevant pieces of information do not come only from the spread at a given time horizon (e.g. 5y) but also from the whole shape of the term structure. For this reason we first introduce the Asset Swap spread over EONIA and then we consider the whole term structure for the countries of interest. This term structure presents also several advantages w.r.t. sovereign CDS spreads and the standard Periphery-Bund spread, avoiding their main limitations.

Finally, once the index has been built, the most challenging task is probably assessing the plausibility of FSI: in [IL06], the authors suggest to compare the computed FSI with the results of experts' description and evaluation of historical stress levels. Unluckily it is documented in the literature that the identification of crisis periods vary from expert to expert (see e.g. [HM05] and references therein). We test our FSI by verifying whether the European Central Bank (hereinafter ECB) responded to the identified episodes of financial stress with an unconventional measure within the next quarter; we also quantify the intensity of ECB response. The empirical literature on ECB reactions to financial instability is rather scant; recently Baxa et Al. [BHV13] have realized an empirical estimate on the monetary policy response to financial instability: they conclude that a pool of central banks that includes the US Federal Reserve and the Bank of England (but excludes the ECB) "often alter the course of monetary policy in the face of high financial stress, mainly by decreasing policy rates" (see [BHV13], pg. 118). Unfortunately their approach is not useful when rates are confined to extreme situations where policy rates are close to the zero bound. In this study we just focus on non-standard policies and on the European case: we come up showing that ECB unconventional measures are strictly related to the events of financial stress indicated by the proposed FSI.

The main contribution of this paper is threefold to the existing literature. First, we introduce and construct the term structure for the Asset Swap spread over EONIA for some government bonds markets. We show that the Asset Swap spread term structure can be modelled by a segmented line with a single breakpoint and we determine the best fit applying the segmented regression technique, a methodology widely used in other fields of research outside econometrics (see e.g. [Hud66; SW03]).
Second, we explain why the shape of the term structure is of particular interest in government bond carry trading, a common trading strategy in peripheral markets even among banking institutions [AS15]. This allows to build an elementary country-index, based on the shape of the whole term-structure and not just on the value of the spread at a given time-horizon.
Third, we propose a basic aggregation scheme for the FSI at the eurozone level. Thanks to the fact that we have a simple, directly observable and intuitive FSI, we can analyse ECB response to this signal with unconventional measures.

The remaining part of the paper is organized as follows. Section 2 is devoted to the description of the country-index, based on the Asset Swap spread over EONIA. In section **3** we identify the episodes of financial stress in the years between 2007 and 2013 and we show how main ECB unconventional measures are related to the financial stress events in the period. Finally in section **4** we state some concluding remarks.

2 A financial stability country-index

As every borrower well knows, sometimes having a continuous access to credit is more relevant than the actual cost paid on the debt. The taboo question that has circulated across the market is the following: is a country like Italy or Spain always able to place its debt?

Let us underline that, for these countries, being able to place their debt is equivalent to having a **seamless access** to the primary market. As an example, we remind that, on average, about €20 billions of Italian debt expire every month. Because this debt is financed with new issues, as it happens for every country, a seamless access to market is crucial "just" to replace expiring debt.

A reasonable index at country level could be

Red:	closed access to primary market (index equal to 2);
Yellow:	restricted primary market access with risk of closing-out (index equal to 1);
Green:	open market access (even if market conditions could remain tough in current economic environment), with the index equal to 0.

In order to estimate from market data the possibility to access to the primary market for a sovereign issuer, in this section we introduce the Asset Swap spread over EONIA; first we show that its term structure can be modelled in an elementary way for the two countries of interest and then we link the access quality to the primary market to the shape of this term structure.

The analysis is carried out considering two extremely liquid markets as the euro government bond secondary market (hereinafter also govies' market) and the Overnight Index Swap (OIS) derivative market. Our dataset covers, over the period between the 1^{st} of January 2007 and the 31^{st} of December 2013, i) all daily last trading prices for Italian BTPs and Spanish Bonos with benchmark size[2] and with expiry within 10 years but longer than 2 months[3] and ii) all euro OIS closing prices. Financial data are obtained from the Bloomberg system.

2.1 Asset swap spread over EONIA as a valuable credit term structure

A Standard Asset Swap contract (S-ASW) in the euro derivate market is described in Figure **1** and involves floating leg payments with Euribor *versus* 3months. The contract is commonly used by a bond investor who desires to hedge its interest rate risk; in conjunction with the bond purchase he enters into an Asset Swap as the payer counterparty, swapping bond coupons with quarterly payments of a 3month Euribor plus a spread. The set of

[2] Issue size larger than €500 mlns.

[3] Shorter bonds are money market instruments and are excluded in the analysis as done by [GSW07] in the US case.

these often simultaneous trades (bond purchase and swap writing) is called an *Asset Swap package*.

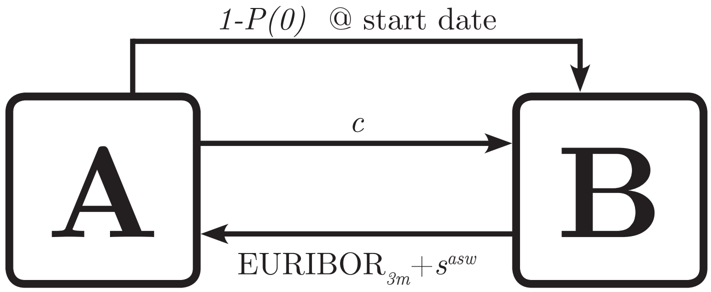

Figure 1: A schematic description of a Standard Asset Swap contract between two counterparties. The payer counterparty (counterparty A in the figure) pays coupons c (either semiannually or yearly for the government bonds of interest) and receives quarterly the 3month Euribor plus a spread (s^{asw} in the figure). Counterparty A also pays at contract start date the difference between par and bond invoice price $P(0)$. In this paper we focus on an ASW where the floating rate, instead of the 3month Euribor, is the 3month OIS rate that "fixes" at the beginning of each quarter.

In this study we consider a new kind of contract: it has the same floating leg dates of a S-ASW, but, in addition to the spread, it also pays – at the end of each 3month lag – the 3month tenor Overnight Interest Swap (OIS) rate that is established at the beginning of the period; the contract is equivalent to a S-ASW, with quarterly floating rate payments, but with an OIS rate instead of a Euribor rate. It can be shown (see e.g. [Hen14]) that such a contract is financially equivalent to an Asset Swap over the EONIA curve with a quarterly paid spread. The spread is called Asset Swap spread over EONIA (hereinafter ASW spread).
The proposed Asset Swap presents two main advantages with respect to a S-ASW: i) it considers the liquid 3m OIS fixing instead of the 3m Euribor rate, whereas a recent scandal has highlighted a certain degree of arbitrariness in the fixing procedure for Euribor rates [CV12] ii) ASW spread corresponds to the credit risk component of an investment in a given issuer, i.e. the additional cost required by the market over a risk free rate (EONIA) in order to hold securities of a given issuer.

The key element in index construction is the term structure (called also the curve) of "Asset Swap spread over EONIA" for Italian and Spanish government bonds. As we have mentioned in the introduction, the relevant information for a policy maker does not come from the absolute value of the spread at a given time horizon (e.g. 5y in [BPS14]) but originates from curve's shape. Several are the advantages of the ASW with respect to standard spread curves already present in the literature as the CDS spread and the Periphery-Bund.
On one hand, ASW spread should be preferred to CDS spread. Sovereign CDS contracts trade mostly with some given expiries (e.g. 10y), moreover they trade very rarely for some countries (e.g. Germany). Furthermore, traded notionals are not publicly available: the

CDS market shows price opacity and it can be manipulated as indicated by the settlement in a recent lawsuit that involved primary dealers and the main info-provider in this market (Markit)[DV15]. ASW spread instead concerns two very liquid markets; moreover, as it is discussed in detail at the end of this section, it is directly related to profits in a carry-trading strategy.
On the other hand, ASW spread at a given time horizon (10y) is similar to the well-known 10y Periphery-Bund yield spread, however there are at least three reasons why the ASW should be the choice to make. First, the EONIA curve is very simple to bootstrap due to the regularity of OIS expiry dates (see e.g. [BC15]). Second, the Periphery-Bund spread overestimates the impact of the additional cost of funds for the peripheral government because it also reflects flight-to-quality effects which tend to reduce Bund yield when an increase in the yield on peripheral securities is observed. Third, ASW spread avoids a technical issue: when the benchmark 10y bond changes either in Germany or in the Peripheral country, a jump in the Periphery-Bund spread could be observed even if the relative credit quality is unchanged.

For both countries of interest, at each value date in the analysed dataset, the curve is obtained by the set of pairs composed by Asset Swap spreads and the corresponding bond expiries[4]. Figure **2** shows the curve in a typical day for Italy and Spain.

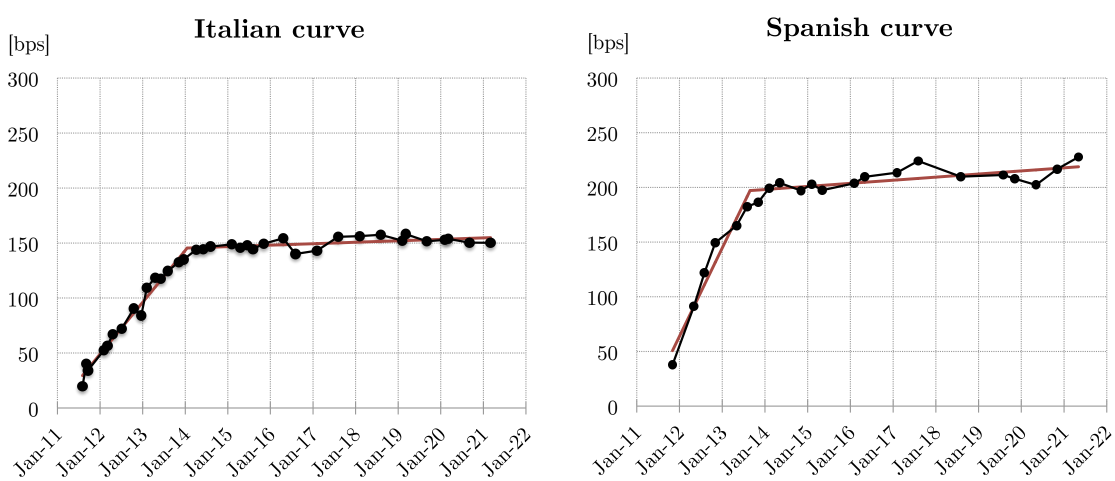

Figure 2: The curve in bps[5] for Italy (left) and Spain (right), on the 1^{st} of June 2011; each point represents the Asset Swap spread of a bond for a given expiry. In red we show the best fit with a single-segmented line.

We observe that the curve can be described by a segmented line with one breakpoint (hereinafter single-segmented line); this technique is presented in the next subsection.

[4]We have applied a simple filter to the observed data as described in the appendix.
[5]One basis point (bp) is equal to 0.01%.

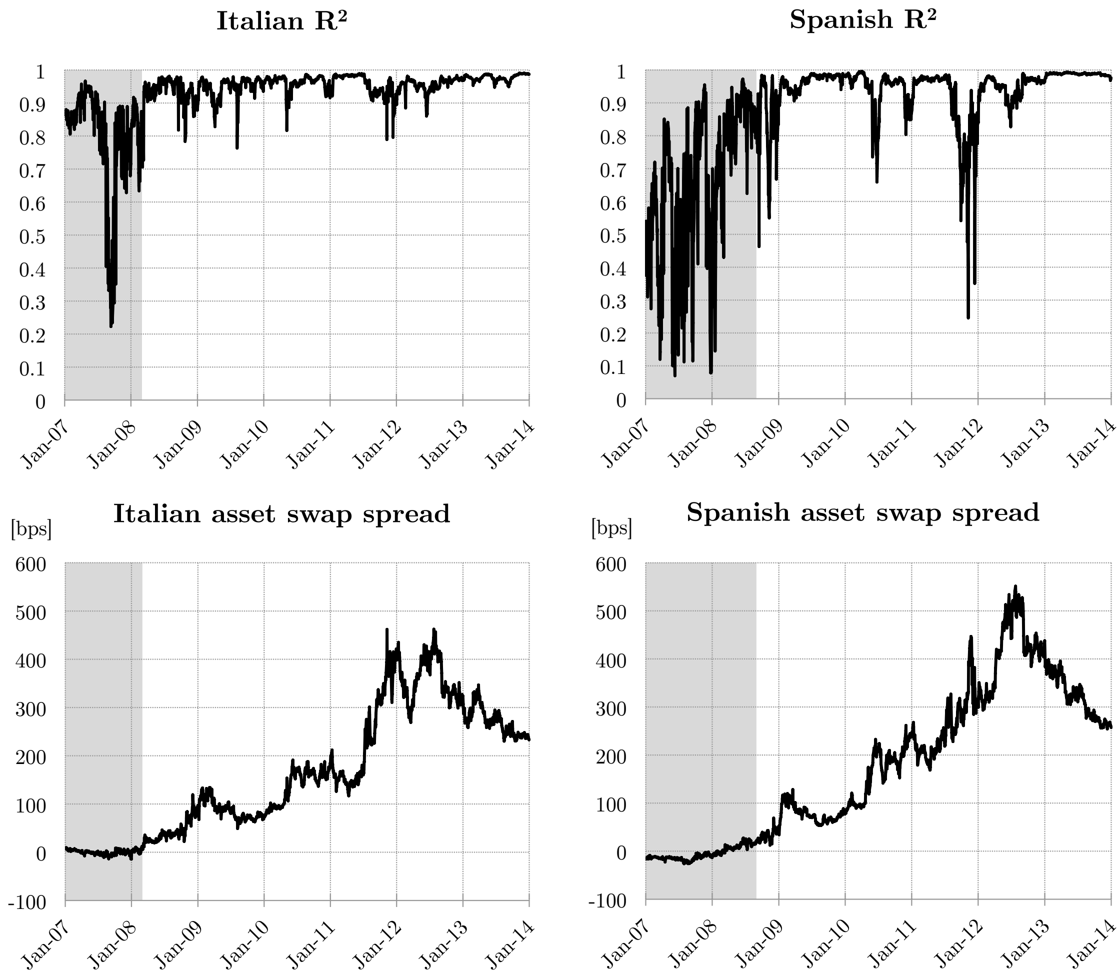

Figure 3: Model coefficient of determination in the Italian and Spanish cases. Comparing with the corresponding values of the 10y-spread, we observe that model description looks adequate outside the "period of relative quiet", i.e. the only time interval where we can hope to see any evidence of financial distress. This period is indicated with a grey shaded colour (up to February '08 for Italy and up to August '08 for Spain).

2.2 Credit term structures as segmented lines

In this subsection we describe how to fit with a single-segmented line the ASW curve: we impose a change of slope with no discontinuity in the time break. We apply an algorithm that reminds [Hud66] in order to find the point of slope change. Details on the technique are reported in the appendix.

This simple model with a single-segmented line for the Asset Swap curve appears to be excellent for the two liquid govies' market under analysis as shown by the coefficient of determination R^2 in Figure **3**. In the same Figure, we have also plotted the value of the 10year-spread (hereinafter also **spread**) for the two countries, marking with a grey shaded colour the "period of relative quiet", that corresponds to the time interval when the **spread** is low, with an average monthly value constantly below 20 bps. It is quite clear

that an average 10year spread lower than such a threshold characterizes a situation where no financial stress could be identified. We observe that outside the "period of relative quiet" the proposed model description looks adequate.
In the Italian case, excluding a first time interval up to February 2008 - the "period of relative quiet" - we observe a R^2 above 0.8 across the period of interest (with the exception of 10 days out of a total of 1495 business days), with an average R^2 equal to 0.96. The situation in Spain is slightly worse, also due to the lower number of bonds available in each value date considered in this study. However, excluding the period with an average monthly **spread** lower than 20 bps (i.e up to August 2008), only 88 values are under the threshold of 0.8 (about 6% of the sample) and only 1% of the spreads have a coefficient of determination less than 0.6, with an average R^2 equal to 0.94.

We call time-to-slope-change the period between the value date and the slope change. After this breakpoint the curve is generally almost constant or with a slight slope on the medium-long term. Usually the curve tends to small values for short expiries; when this condition is not verified the curve is called "inverted" and signals that the market is evaluating carefully issuer credit quality and it considers a default likely in the short term. We recall that our purpose is not to obtain an indicator of the market access quality for an issuer on a single day but a signal on a time period of particular relevance for a policy maker, i.e. the month. For this reason it is useful to consider the mean time-to-slope-change over a month: this average time-to-slope-change (hereinafter called **time**) is a relevant quantity for a bond issuer as we discuss in detail in the next subsection.

The presence of a slope change in the curve makes possible to identify two other relevant indicators, beyond the 10-year **spread**, the typical reference already known also by the man of the street: the **time** and the (sign of the) slope before the break-point (**slope**). With these three indicators we get a simplified but accurate description of the curve.

2.3 Financial interpretation of indicators and the role of carry trade

The financial meaning of two indicators is rather simple and well known. The **spread** is the cost demanded by the market in addition to a security considered risk-free in order to remunerate the risk undertaken by 10y bondholders of the issuer of interest while the (sign of the) **slope** indicates whether the sovereign market has a normal term structure (positive sign) or an inverted one (negative sign). An inverted **slope** signals that the access to primary market is closed: the market believes the issuer is either near to default or about to enter into a restructuring plan for its debt.

The other indicator – **time** – needs a more detailed explanation related to the behaviour of a government debt carry-trader, a typical investor in peripheral euro govies' markets. This kind of investor normally buys a peripheral bond and, at the same time, he immunizes himself from interest rate risk by writing as a payer counterparty an Asset Swap contract with the same expiry; i.e. he enters into an *Asset Swap package* also called a *package*. A carry-trader builds his investment strategy having in mind a *static* scenario, in which the Asset Swap spread curve does not vary during time.

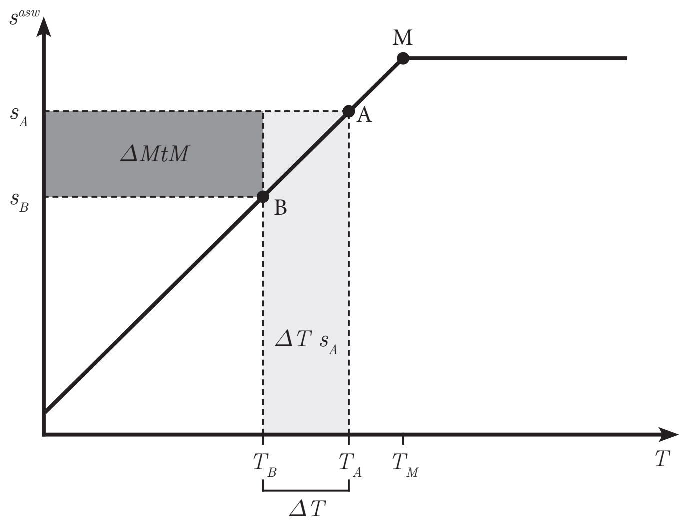

Figure 4: Typical curve spread s^{asw} observed in the market as a function of time horizon: the term structure is well approximated by a straight line with a positive slope up to a point M (time-to-slope-change); after M the slope becomes close to zero. We describe also some features of the carry-trade strategy in a govies' market, if a carry trader enters into a *package* with expiry T_A and he closes the position after a lag ΔT. In a *static* scenario, this strategy originates two sources of profit - represented by two rectangles of different colours - one related to cash balances (light grey) and the other to the difference in Mark-to-Market (dark grey), as described in the text. In the situation described in the figure, the point of slope change M represents the expiry of the *package* for which the carry is maximized.

In the financial literature it is often believed that "carry trade reflects a bet that eurozone countries would converge economically, resulting in a convergence of the spread [curve]" (see e.g. [AS15], page 216). Clearly, in a trading strategy that involves *Asset Swap packages*, the profit can result from a shrinkage in the spread curve: in this case we are dealing with a directional component of the profit; it can always derive from either a long or a short position on an asset. However, it can be useful to stress that carry arises just considering a *static* scenario. In the marketplace carry trade profit identifies the amount earned in the strategy *rebus sic stantibus*, just consequence of having that given position and keeping it for a lag ΔT, not due to a change in market conditions, i.e. a shrinkage of the spread curve in this case.

For this reason, let us consider the two components in the Profit&Loss (P&L) of this trading strategy after ΔT in a *static* scenario as described in figure **4**: i) cash balances, i.e. cash flows received/paid after the settlement date t_0[6] entering into a *package* with expiry T_A and keeping it for ΔT (including the cost-of-carry) and ii) the difference in Mark-to-Market (MtM) for the *package* in the lag between the opening and the closing of the carry position.

In a *static* scenario, and then in absence of default, the *package* with expiry T_A is equivalent to a floater [Sch03] with a spread over EONIA s_A. Moreover, short term funding costs for

[6]The settlement date t_0 is equal to two days after the value date.

eurozone govies (i.e. Repo rates) are well approximated by EONIA rate. Once funding costs have been included, keeping such a floater with expiry T_A for a lag ΔT generates cash balances equal to the Asset swap spread s_A relative to the period ΔT.
In the appendix we show in detail each component in the P&L report for a carry trader in the *static* scenario; in this section we describe the typical observed term structure in the market, shown in figure **4**, with a positive slope present up to the time-to-slope-change T_M and an ASW curve flat afterwards. This case allows to understand qualitatively what are the key elements for a carry trading strategy in the government bond market. Figure **4** outlines a simplified situation where i) we neglect discount factors, ii) we omit the exact value of the slope after the time-to-slope-change and iii) we consider continuous payments in the floating leg of the swap.
In this simplified case, cash balances are given by $s_A \cdot \Delta T$ (represented in Figure **4** as a light grey rectangle). This component in the carry trade is rather intuitive and well-known (see e.g. [AS15]). The other component of P&L, i.e. Mark-to-Market, is however in most cases a significant part of profit. In figure **4** it is shown that, for T_A lower or equal T_M, a MtM component in P&L is observed; it is equal to the difference between the two spreads, s_A and s_B, multiplied by the remaining time-to-expiry for the *package* T_B (dark grey in the figure). As a matter of fact, in the described scenario, after ΔT a new *package* can be traded on the market with a spread s_B lower than s_A: from the sale of the owned package the carry-trader will collect a price higher than the initial one. This additional price equals the difference in value between two floaters of the same issuer with same expiry and two different spreads s_A and s_B, i.e., in the simplified situation described in the figure, $\Delta MtM = T_B \cdot (s_A - s_B)$.
Summarizing, a carry-trader buys a *package* with an expiry in T_A and he holds the package for a given interval $\Delta T = T_A - T_B$ (so the package, after ΔT, has a time-to-expiry T_B). In the *static* scenario described in figure **4** the carry-trader gains a double profit: on one hand, he assures himself the extra return given by Asset Swap spread roughly equal to $\Delta T \cdot s_A$; on the other hand, he can sell the package at a higher price after ΔT.
Clearly, among all available packages for the country of interest, a carry-trader will try to buy the one which maximizes his carry, i.e. his profit in a given time interval ΔT. As shown in Figure **4**, the carry-trader reaches his goal of profit maximization when he buys securities with expiry on T_M, the time-to-slope-change.

This indicator is a relevant quantity in govies' markets: on one side, it is the time horizon of particular interest for a typical investor in a government bond market and, on the other, the time horizon up to which an issuer finds easier to place its bonds, due to an active demand.

As already mentioned, we build our financial stress index first considering FSIs at country level and then aggregating them at eurozone level. We are interested in an indicator for a policy maker, this is the motivation for considering the monthly average of the time-to-slope-change (hereinafter **time**). In country-FSI we compare this **time** with a given threshold. We claim that carry-trading is a key strategy in the euro govies' market and it has been observed that "the majority of this bond purchases had maturity of three years or less" (see e.g. [AS15], page 220). For this reason, the threshold in the country-FSI has been chosen equal to two years since, on one hand, three years is the horizon up to which the carry trading mechanism is more pronounced and, on the other hand, one year, the time horizon regulated by country's Finance Act, is the minimum credibility period for a country. Two years appears to be a situation where a sovereign is facing some serious difficulties when placing its debt: in the next section we also verify the robustness of the results with a different threshold selection.

At this point, the FSI at country level can be easily constructed: when **slope** is inverted, a severe market disruption is taking place (Red), if **time** decreases below the 2-year threshold we have a signal of dysfunctional market (Yellow), otherwise we observe an open market access (Green). More precisely, a month has a Yellow light if **time** is below two years and contemporarily an increase of the **spread** is observed.[7] Light turns to Red in months with a negative **slope** for more than 5 consecutive business days (i.e. 1 week) and, at the same time, an average monthly spread higher than 100 bps.

The FSI at eurozone level can be obtained noticing that it is enough to find one large country without access to financial markets for a collapse of the euro: as already discussed in the introduction, systemic risk arises from the danger of a euro-area break up. We have to include only "large" countries in the FSI; a small country can overcome its difficulties via direct lending by other countries or via financial assistance by EU mechanisms (e.g. ESM): this possibility is not viable if a large country is involved. The FSI at eurozone level is then defined as

$$FSI = \max_{i \in Large\ Countries} FSI_i \quad . \tag{1}$$

In this way the aggregation step is immediate and avoids most of the arbitrariness in the determination of weights in standard FSI.

3 Identifying episodes of financial distress

The key goal of this study is building a simple FSI in the euro area and relating it to ECB unconventional measures. First we briefly recall the main ECB unconventional measures and we construct our country indicators for Italy and Spain identifying the episodes of financial distress and their severity, related to the possibility of having a seamless access to the primary market of these two countries. The aggregate indicator at the euro area level is just the maximum of the indicators obtained at country level. ECB unconventional measures offer a unique opportunity for verifying the quality of the financial stress signal produced by the proposed FSI: at the end of this section we analyse the link between the signalled crisis episodes and ECB unconventional measures.

3.1 ECB unconventional measures

Due to the abundance of taxonomies currently available in the literature we briefly summarize in Table 1 the six main unconventional measures put in place by ECB.

[7] We require that the value of end-of-month **spread** is higher than the beginning-of-month **spread** on previous Green month, i.e. previous month if it is Green, otherwise the last previous month with a Green colour. As previously mentioned, in order to observe a financial stress situation, we always need to be outside the "period of relative quiet", i.e. that the average monthly spread is higher than 20 bps; if spread is lower than 20 bps we cannot consider that month a period of financial stress.

Acronym	*Description*	*Date*	*Amount*
ECS-1	Enhanced Credit Support (Part 1)[8]	Oct '08	unquantifiable
ECS-2	Enhanced Credit Support (Part 2)[9]	May/Jun '09	€470 bln
SMP-1	Securities Market Program (Part 1)[10]	May '10	€70 bln
SMP-2	Securities Market Program (Part 2)	Aug '11	€140 bln
LTRO	3y-Long Term Repo	Dec '11/Feb '12	€1030 bln
OMT	Outright Monetary Transactions[11]	Aug '12	unused

Table 1: The six sets of unconventional measures activated by ECB, with the date of announcement/realization and the amount involved.

The first two sets of non-standard measures are called Enhanced Credit Support, after the well-known speech of ECB President at the University of Munich [Tri09], and they focus mainly on the banking system, aiming at ensuring the transmission of policy rates. The Enhanced Credit Support (ECS) consists in a host of unconventional measures that can be divided schematically into two sets. The first part of these measures (ECS-1) was announced by ECB on the 15^{th} of October 2008 [ECB08], one month after Lehman's default and involved: i) unlimited provision of liquidity through fixed rate tenders with full allotment, ii) extension of the list of collateral assets and iii) liquidity provisions in US dollars through FX swaps with Federal Reserve. With the second part of the program (ECS-2), announced on the 7^{th} of May 2009 [ECB09], the 12month Long Term Refinancing Operation (12m-LTRO) and the Covered bond purchase programme have been created. This last measure had an impact on the Eurosystem's consolidated balance sheet of less than €500 bln.[12] On the 10^{th} of May 2010 ECB announced its third unconventional measure [ECB10b], the Securities Market Program (also known as SMP-1), a sovereign bonds purchase on the secondary market of the three smaller peripheral countries' bonds (Greece, Ireland and Portugal), with a total purchase of about €70 billions. This program was re-opened [ECB11a] on the 2^{nd} of August 2011 (SMP-2) and, as recent research has pointed out [Fra12], SMP-2 was mainly focused on Italian and Spanish bonds.[13] On the 8^{th} December 2011 ECB announced its fifth unconventional measure [ECB11b] the two 3-year Long Term Refinancing Operations (3y-LTRO) that took place on the 22^{nd} of December 2011 and on the 29^{th} of February 2012 for a total amount of about €1.03 trillions; we would like to mention that this is the only unconventional measure above €1 trillion deployed in the euro area, i.e. of the same order of magnitude of Italian and Spanish debt.[14] Our

[8] Unlimited liquidity provision through fixed rate tenders with full allotment, extension of the list of assets eligible as collateral, US dollar liquidity provision through FX swaps.

[9] 12month LTRO and Covered Bonds Purchase Program.

[10] Measure in conjunction with the Greek Loan for €110 bln by EU/IMF.

[11] Measure in conjunction with the financial assistance to Spain announced by EU for €100 billions and realized via the ESM that disbursed a total of €41.3 billions for the recapitalization of country's banking sector.

[12] The total amount allotted in the first 12m-LTRO on the 24^{th} of June 2009 was €442 bln.

[13] We have divided SMP in two parts, since, for all practical purposes, we are dealing with two different policy maker interventions [Fra12].

[14] Even if the exact amount involved by the first unconventional measure (ECS-1) is difficult to estimate,

main criteria to determine the relevance of an unconventional measure is related to the impact on the Eurosystem's consolidated balance sheet, since, as suggested by Trichet, "the appropriate metric for measuring the non-conventional monetary policies is the increase of the size of the balance sheet of the central bank which is due to outstanding monetary policy operations" (see [Tri13], pg. 241).
ECB announced on the 2^{nd} of August 2012 the intention to perform Outright Monetary Transactions (OMT) [ECB12]: this is the last ECB unconventional measure and it has remained unused in the period analysed in this study.

3.2 Measuring indicators at country level

The three indicators described in subsection **2.2** (**time**, **slope** and **spread**) for Italy are shown in Figure **5**.

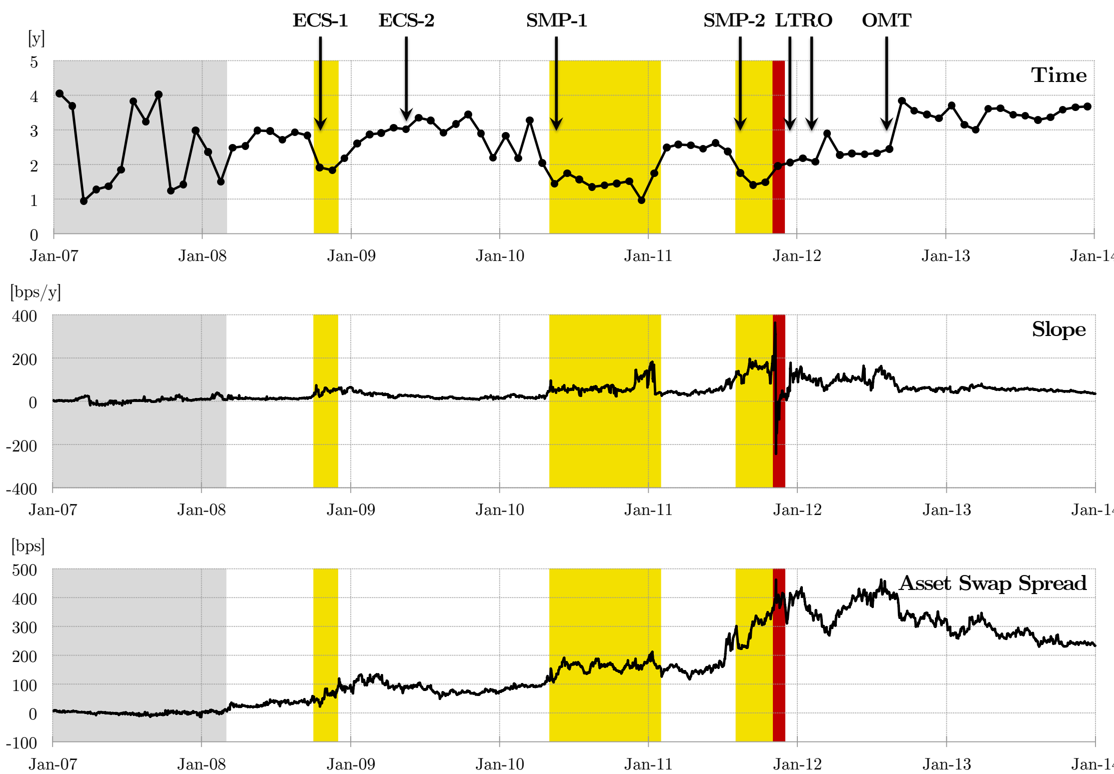

Figure 5: Time evolution of the three indicators for Italian BTPs. We indicate with a grey shaded colour the time interval characterized by an average monthly spread constantly below 20 bps (up to February '08), called "period of relative quiet" in the text. The three zones of market turbulence are identified.

Figure **5** displays three areas where **time** is below the 2year threshold. We observe, comparing the graphs, that **slope** and **spread** grow in the highlighted zones. These three zones identify three particular events that have influenced credit access for the Republic

in any case it is well below €1 trillion.

of Italy. The default of Lehman Brothers, during September 2008, first questioned world financial stability. Later on, during May 2010, the Greek crisis showed – for the first time – the unsustainability of the debt for a European country; throughout this period we note the growth of **spread** and the simultaneous decrease of **time** under two years. This period of uncertainty lasts till the end of the year, with **time** that overtakes the 2year threshold only in February 2011. **Time** falls under the 2year threshold during August 2011, again in conjunction with a growing spread: ECB decides promptly to extend the SMP programme to Italy and Spain. We observe that, following the **slope** inversion in November 2011, 3y-LTRO was announced and executed: this is the most significant unconventional measure in the euro area at least (but not only) in terms of volumes involved.

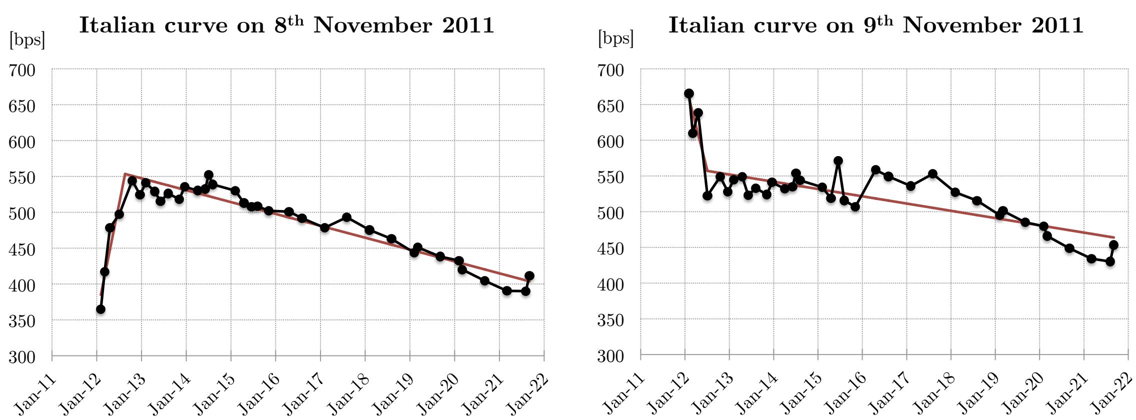

Figure 6: We compare Asset Swap spread curve, in bps, for Republic of Italy on the two consecutive days, the 8^{th} and 9^{th} of November 2011. We observe that inversion occurs abruptly, from one day to the other.

It is interesting to observe the curve in the day when the inversion of **slope** took place for Italy, comparing it with previous day values, as shown in Figure **6**: investors didn't buy new bonds but they tried to sell the ones they held, perceiving the possibility of an immediate default. Let us stress one of the main results of this study: relevant market information comes from analyzing the full shape of spread curve instead of considering just the value of the 10y-spread (a too-coarse scalar index).

Results for Spain are shown in Figure **7**. As in the Italian case **time** goes below the 2year threshold for the first time after Lehman's default. The so called Greek crisis is signalled both at the beginning (in May and June 2010) and at the end (from November 2010 to January 2011); in the Spanish case, just after the SMP-1 unconventional measure, **time** goes above the 2year threshold. A financial stress is signalled again for the major-GIPSI crisis on the second half of 2011; also for the Kingdom of Spain we observe a slope inversion in November 2011. The last crisis period is the one related to the so-called Spanish banking crisis that was followed by the announcement of the OMT program.
It is probably useful to observe that this last crisis is not indicated by our country-index for Italy, even if the 10y spread was approximately at the level reached during the apex of the crisis on previous November. However, in the first quarter of 2012 the Italian situation was normalizing with a technocratic government lead by Mr Monti and the proposed index

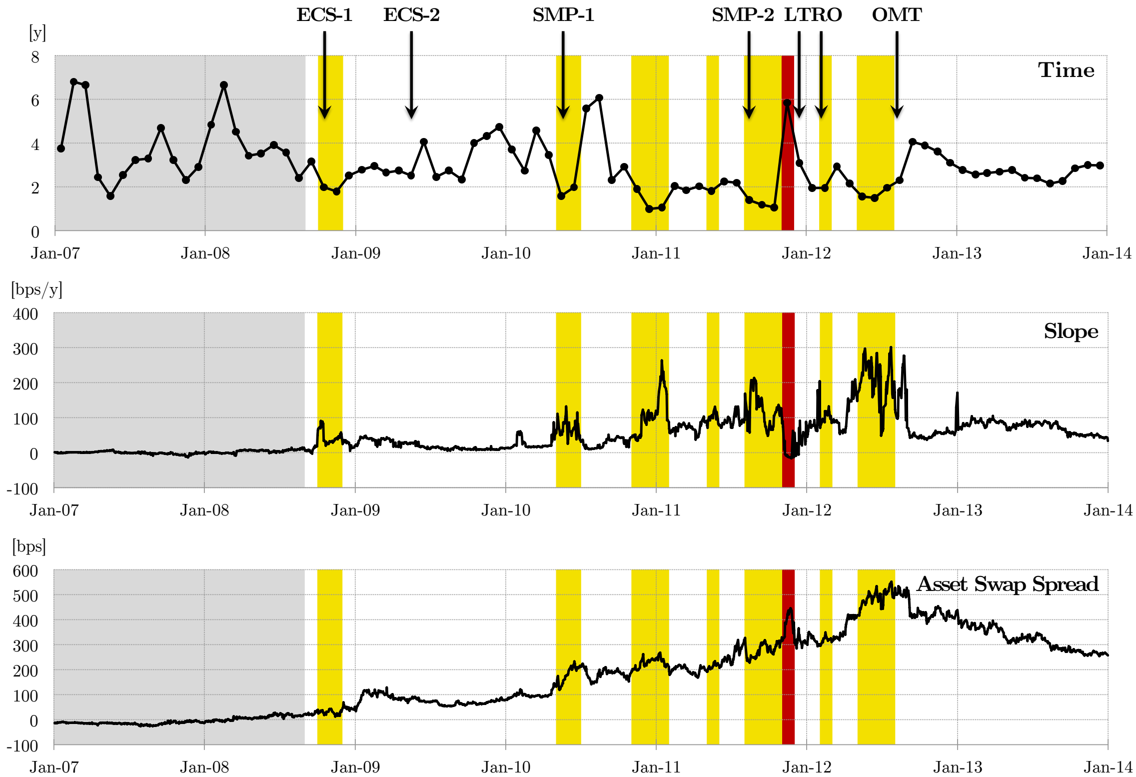

Figure 7: Time evolution of the three indices for Spanish government bonds. We indicate with a grey shaded colour the period characterized by an average monthly spread constantly below 20 bps (up to June '08); we also identify the zones of market turmoil with yellow and red colours.

reveals that the observed spread movement is due to contagion. Clearly no indicator based uniquely on the value of the 10y spread can distinguish between the two situations and correctly select the dysfunctional market event.

In this study we have also tested the robustness of our results in several directions. First, we have analysed the case of France, the other "large" country in the euroarea. A yellow signal is observed only on January 2009, the first month after the "period of relative quiet" for this country; no other situation of financial distress is signaled in the period under analysis and then, as one could expect, this country does not provide any contribution to the FSI defined in equation (1).
We have also considered different possible choices for the parameters in the FSI. Clearly the thresholds for the 10year-spread can be chosen in quite a large range and the results do not change. For example choosing the threshold for "the period of relative quiet" in the range 10 to 30 bps, one obtains exactly the same results. More delicate is the situation for the threshold on **time**: the two year threshold is a critical value for **time** chosen in the range between one year, the minimal credibility period for a sovereign issuer, and three years, the most relevant period for carry trades. One can also require not only a condition on the average time-to-slope-change (i.e. **time**) but also taking into account a confidence level. As confidence level we have considered monthly standard deviation σ of time-to-slope-change. We have tested that the sum of **time** and two σ being lower

of three years. Results are qualitatively similar. For Italy the proposed FSI signals the same stress periods as before and the forth yellow zone due to banking Spanish crisis; for Spain there is just one period more - the months of March, June and July 2013 - w.r.t. the ones obtained with the proposed FSI: this additional period is simply due to the low volatility in the credit term structure observed for Spain (and then a small σ), due to a stabilization of the sovereign debt crisis after the announcement of OMT.
The results we obtain are robust: an additional advantage given the simplicity of the proposed FSI.

3.3 ECB response to European financial stress events: unconventional measures

Figure **8** shows how the suggested FSI works and it is compared with ECB unconventional measures. Each month is identified with a colour, for both Italy and Spain. We observe that the intervention of the policy maker follows the start of a yellow area for one of the two countries.

Figure 8: Zones of restricted primary market access (Yellow) and severe market disruption (Red) highlighted by the proposed FSI.

After the beginning of a stress event indicated by the suggested FSI, ECB adopts an unconventional measure within the next quarter, the typical intervention lag for a policy maker. In particular after the red area, ECB implemented the 3y-LTRO: the situation fully justified the relevance of this measure.
The only exception to this pattern in the use of unconventional measures is the ECS-2, announced on May 2009. In this case our opinion is that ECB overreacted to the banking crisis that followed Lehman's default, probably due to the fact that it was operating in uncharted territories. The apex of the crisis was probably over: on the 2^{nd} of April, during the G20 Leaders' Summit on Financial Markets and the World Economy, the leaders pledged more than 1 trillion dollars to tackle the global crisis and all equity markets started a positive trend.

We have shown that crisis events have been followed by ECB unconventional measures, their timing and intensity – together with a smart communication – being the source of their effectiveness. Credibility of ECB unconventional measures is a two-sided story. On one side we have the message: ECB "is ready to do ***whatever it takes*** to preserve the euro. And believe me, it will be ***enough***" [Dra12]. On the other side we have the prompt response with an unconventional measure each time a market dysfunction (Yellow market signal) or even a severe market disruption (Red) takes place.
In this study we show that ECB response is simple and unambiguous with an actual alignment between words and deeds: each time a market light turns to Yellow, ECB is

ready to put in place an unconventional measure (***whatever it takes***) and the most relevant initiative is activated (***enough***) when the Red light is on.

4 Conclusions

In this paper we have suggested a new Financial Stress Index for the euro area economy. We have shown how to build a simple discrete market-based FSI which identifies the episodes of systemic stress and which signals the level of instability.
The index is based on some features and on the shape of the credit term structure for the "large" countries in the area. This method has the advantage of an overwhelming simplicity, both in terms of index components selection and aggregation, due to some specific characteristics of the euro area economy.

There are several significant innovations with respect to the existing literature.
First, we have focused the analysis on the curve of Asset Swap spread over EONIA, a credit term structure that turns out to be more accurate than CDS spreads and the well known Periphery-Bund spread. In particular we have shown that this term-structure can be modelled very accurately via a single-segmented line and we have underlined that the main information content comes from the specific shape of the spread curve more than from absolute changes in yields.
Second, we have discussed in detail how the particular shape of the term structure of Asset Swap spreads is related to the behaviour of a government bond carry trader.
Third, we have built a country-index that focuses on country access to the primary debt market. The country index involves, besides the 10y-spread, two other key financial distress indicators: the **time** and the (sign of the) **slope**. The proposed FSI is just the maximum of the country-indices: a simple aggregation step that avoids most of the arbitrariness in the existing FSIs.
Finally, we have shown that, after every stress event indicated by the suggested FSI, ECB adopts an unconventional measure; its most relevant initiative - the 3y-LTRO - has been announced and executed immediately after the occurrence of the only severe market disruption in the period under analysis.

The proposed Financial Stress Index is based on systemic relevance of credit term structure for some key countries in the euro area and it reflects market participants' behaviour. This FSI looks promising, as suggested by the empirical results in this study: despite its simplicity and immediacy it signals the major financial crises in the area very precisely and in a robust way, identifying not only the most severe episodes of stress in the sovereign debt crisis but all periods of financial instability in the analysed period, including the banking crisis that has followed Lehman's default.

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Appendix

Asset Swap spread

In this paper we consider the "Asset Swap over EONIA"; this contract is equivalent to a Standard Asset Swap, with quarterly floating rate payments in arrears and fixings in advance, but with an OIS rate instead of a Euribor rate. The contract has, in general, a short stab in advance; the OIS rate for this shorter lag is the one with a tenor equal to the corresponding period.
For each value date in the dataset analysed, we consider all bonds tradable on the secondary market for the issuer of interest and we compute Asset Swap spreads as a function of bond expiry dates. The spread is obtained requiring that the Net Present Value of the Asset Swap is equal to zero at value date. After simple algebra, in the standard modelling framework (see e.g. [Hen14]), it follows that

$$s^{asw} = \frac{1}{\sum_{k=1}^{K} \delta(t_{k-1}, t_k) B(t_0, t_k)} \left[B(t_0, T) - P(0) + c \sum_{n=1}^{N} \delta^f(t^f_{n-1}, t^f_n) B(t_0, t^f_n) \right]$$

where

$P(0)$:	bond invoice price at the value date;
c:	bond coupon rate;
t_0:	settlement date, two business days after value date;
$\{t^f_n\}_{n=1,..,N}$:	bond coupon dates, i.e. fixed leg payment dates;
$\{t_k\}_{k=1,..,K}$:	quarterly (modified foll.) payment dates in the floating leg of the ASW;
$T = t_K = t^f_N$:	bond (and Asset Swap) expiry date;
$\delta_k \equiv \delta(t_{k-1}, t_k)$:	floating leg year fraction between t_{k-1} and t_k ($Act/360$ daycount);
$\delta^f(t^f_{n-1}, t^f_n)$:	coupon leg year fraction for the lag (t^f_{n-1}, t^f_n) with coupon daycount;
$B(t_0, t)$:	discounting curve (EONIA) between t_0 and t.

Carry with an Asset Swap package

In this appendix we detail - in a *static* scenario - the P&L of a carry trade position opened in t_0 and closed in t_c, after a lag ΔT, with funding costs at EONIA. We consider the long position in the *package* - long bond and payer counterparty in the Asset Swap - in the situation of perfect hedge of the interest rate risk, i.e. at the beginning of each quarter the carry trader writes also a 3m OIS contract[15] as receiver counterparty for the same notional of the *package*. Let us suppose that there are n (floating leg) reset dates between t_0 and t_c. Taking into account the funding costs involved in this trading strategy, P&L in

[15] More in general the receiver OIS is up to the next reset date in the floating leg.

t_c is equal to

$$+\left[1+\delta(t_n,t_c)L(t_n,t_c)\right]\sum_{i=1}^{n}\delta_i s_A\prod_{j=i+1}^{n}\left[1+\delta_j L(t_{j-1},t_j)\right] \qquad \text{spreads in the } \textit{package} \text{ up to } t_n$$

$$+\left[1+\delta(t_n,t_c)L(t_n,t_c)\right]\sum_{i=1}^{n}\delta_i L(t_{i-1},t_i)\prod_{j=i+1}^{n}\left[1+\delta_j L(t_{j-1},t_j)\right] \qquad \text{floating rates in the } \textit{package} \text{ up to } t_n$$

$$+\delta(t_n,t_c)s_A B(t_c,t_{n+1}) \qquad \text{last spread before } t_c$$

$$+\delta(t_n,t_c)L(t_n,t_c) \qquad \text{last floating rate before } t_c$$

$$-\left[1+\delta(t_n,t_c)L(t_n,t_c)\right]\prod_{j=1}^{n}\left[1+\delta_j L(t_{j-1},t_j)\right] \qquad \text{cost in } t_0 \text{ \& cost of carry}$$

$$+1 \qquad \textit{package} \text{ sale price in } t_c$$

$$+\delta(t_c,t_{n+1})(s_A-s_B)B(t_c,t_{n+1}) \qquad \text{first term in MtM}$$

$$+(s_A-s_B)\sum_{i=n+2}^{K}\delta_i B(t_c,t_i) \qquad \text{other terms in MtM}$$

where $L(t_{j-1},t_j)$ is the variable rate taking into account compound overnight EONIA interest in the period between t_{j-1} and t_j as defined in equation (1) in [Eur08] and the empty product is equal to 1. The terms can be easily identified: in t_c the trader sells a "new" *package* at par, and then with a spread s_B.

After simple algebra, from the above expression one gets the two components in P&L discussed in section 3

$$\left.\begin{array}{r} +s_A\left[1+\delta(t_n,t_c)L(t_n,t_c)\right]\displaystyle\sum_{i=1}^{n}\delta_i\prod_{j=i+1}^{n}\left[1+\delta_j L(t_{j-1},t_j)\right] \\ +s_A\,\delta(t_n,t_c)B(t_c,t_{n+1}) \end{array}\right\} \begin{array}{l}\text{cash balances}\\ \text{capitalized up to } t_c\end{array}$$

$$\left.\begin{array}{r} +(s_A-s_B)\,\delta(t_c,t_{n+1})B(t_c,t_{n+1}) \\ +(s_A-s_B)\displaystyle\sum_{i=n+2}^{K}\delta_i B(t_c,t_i) \end{array}\right\} \text{MtM discounted in } t_c$$

Filtering technique

Our database covers all daily last trading prices for Italian BTPs and Spanish Bonos in the period between the 1^{st} of January 1999 and the 31^{st} of December 2013. The selected bonds pay fixed coupons, they have an outstanding higher than the benchmark size, and expiry date within 10 years but longer than 2 months: 81 Italian BTPs and 45 Spanish Bonos match the criteria, with an average of 34 BTPs and 21 Bonos every value date.

This dataset has been filtered in order to eliminate unrealistic values. For a given bond it is possible for the Asset Swap spread to present a large jump (in absolute value more than a threshold of 50 bps) from one day to the following business day, and again, the following day another significant jump (more than the threshold) but with the opposite sign. These large up-and-down (or down-and-up) movements from one day to the other seem related to ephemeral events, probably tied to market illiquidity of some last traded prices. Whenever it happens, the "central" value in the triplet has been filtered out and replaced with the average spread between the first and the third values of these three business days intervals. The number of values modified by this filter is negligible: only 19 out a total of $58,230$ values for BTPs and 4 out of $37,157$ Bonos values have been affected by the filter.

The algorithm to fit with a single segmented line

The algorithm to detect the optimal single segmented line and to choose the optimal regression parameters is elementary and involves only independent and constrained linear fits [Hud66]; its scheme is described in Figure **9**.

For each business day we have a sequence of n pairs describing the curve: the Asset Swap spreads $\{s_i\}_{i=1,..,n}$ and bonds' expiries $\{T_i\}_{i=1,..,n}$. We iteratively split the sequence in two sub-intervals, one containing the first k points and the second containing the last $n-k$ points, with $k=3,\ldots,n-3$; at step k we compute a time-break candidate τ_k and L_k, the candidate for the least residual sum of squares between the original data and the fit proposed. First, two independent linear regressions are computed on each sub-interval. If the two regressions return two different sets of parameters the two straight lines have an intercept. If they meet inside the interval included between the k^{th} (included) and the $k+1^{\text{th}}$ expiries, then the point detected is the k^{th} candidate point of time-break. Otherwise, the algorithm tests both extreme expiries (t_k and t_{k+1}) as abscissa of the change point.[16] Each extreme point is considered a candidate point of time-break and two linear regressions are performed, constrained by the continuity on the time-break. After the detection of all candidate points of time-break, one determines k^* the index that corresponds to the least sum of squares $L^*_{k^*}$ among value candidates and then to the time-to-slope-change $\tau^*_{k^*}$.

It is clear from the above algorithm description, that the segmented regression selects either an independent linear fit on two disjointed sets of points or a linear fit with a linear constraint on the break-point. Contrary to [Hud66] the above algorithm requires that each one of the two sets of pairs contains at least 3 points in order to have non-trivial fits.

Whereas segmented regression is a simple generalization of the standard linear regression, some properties of the coefficient of determination R^2 hold, where we define the coefficient of determination R^2 as in the standard linear case

$$R^2 \equiv 1 - \frac{L^*_{k^*}}{TSS}$$

[16]As explained in [Hud66], since the residual sum of squares in the constrained case is always greater or equal to the independent case, this test is performed only if the residual sum of squares for the two independent regressions is lower than all candidate values $\{L_j\}_{j<k}$ computed up to step k.

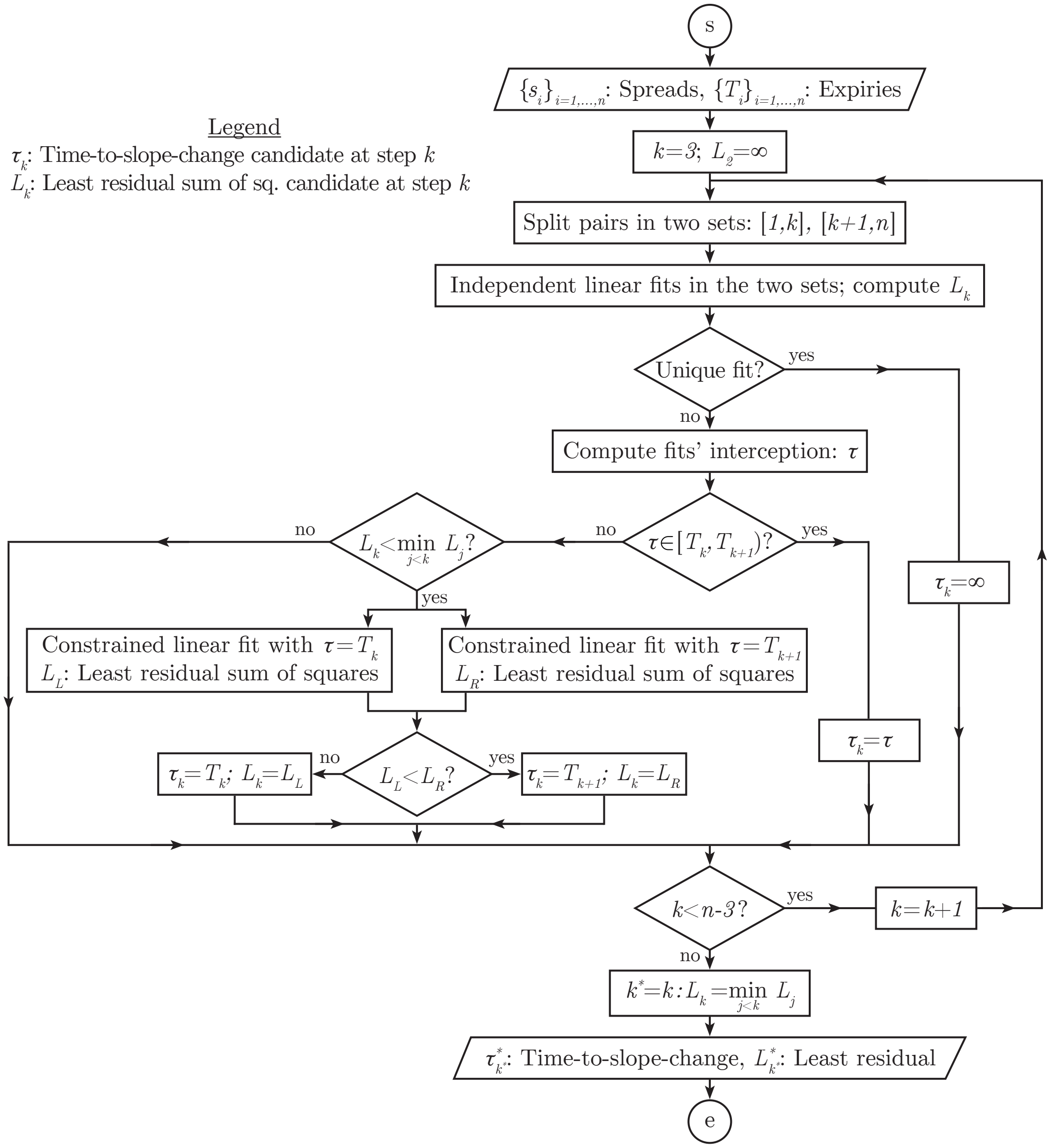

Figure 9: The scheme of the algorithm for segmented regression. It determines the optimal single-segmented line and the parameters of the regression.

with

$$TSS \equiv \sum_{k=1}^{n} (s_i - \overline{s})^2$$

being the Total Sum of Squares and $\overline{s}$ the average Asset Swap spread.
Also in this case the coefficient of determination is an elementary indicator about the quality of model fit. For example two properties that hold are: i) when R^2 equals 1 the segmented regression perfectly fits the data and ii) $R^2 \in [0,1]$ as stated in the following lemma.

Lemma
In segmented regressions the coefficient of determination R^2 ranges from 0 to 1.

Proof: The upper bound is immediate since both $L^*_{k^*}$ and TSS are positive numbers. The lower bound is a natural consequence of the following property: as shown in Figure **9**, the algorithm could return a standard linear fit with no structural change. Defining L_{linear} the least residual sum of squares in this linear case, we clearly have that this value is greater or equal the least residual sum of squares in segmented regression case, i.e. $L_{linear} \geq L^*_{k^*}$, and then $L^*_{k^*} \leq L_{linear} \leq TSS$. ♣